MODELI PODATAKA U ELEKTROENERGETSKIM SISTEMIMA

VEŽBA 2: Upotreba CIMTool alata za kreiranje CIM profila u RDFS formatu

CIM, RDF, RDFS

- CIM (Common Information Model)
  - Apstraktni model koji je razvila Radna grupa 13 Tehničkog komiteta 57 Međunarodne elektrotehničke komisije
  - Postao je međunarodni standard IEC61970-301.
  - Opisan je pomoću UML-a.
- RDF(ResourceDescriptionFramework)
  - Definisan familijom W3C (World Wide Web Consortium) standarda kao model za opis mašinski čitljivih meta-podataka.
  - Opisuje resurse u obliku subjekat-predikat-objekat “tripleta”
- RDFS(ResourceDescriptionFrameworkSchema)
  - Proširivi jezik koji obezbeđuje elemente za zadavanje RDF resursa.

CIM PROFILE

- Profil predstavlja skup definicija klasa njihovih atributa I međusobnih veza u okviru jedne šeme. Profil predstavlja opis modela podataka koji je podskup neke nadređene šeme.
- Osnovna uloga profila je definisanje domenskih i kontekstno zavisnih modela.
- Zadavanje profila je moguće u nekoliko oblika i to RDFS XSD kao tekstualni ili HTML document.
- Jednostavno rečeno svi delovi CIM a su opcioni u zavisnosti od aplikacije i njenih zahteva. Zbog ovakvog stava neophodan je bio poseban dokument koji bi specifikovao neophodne delove. Takav dokument je nazvan **CIM Profil**.

CIMTOOL ALAT

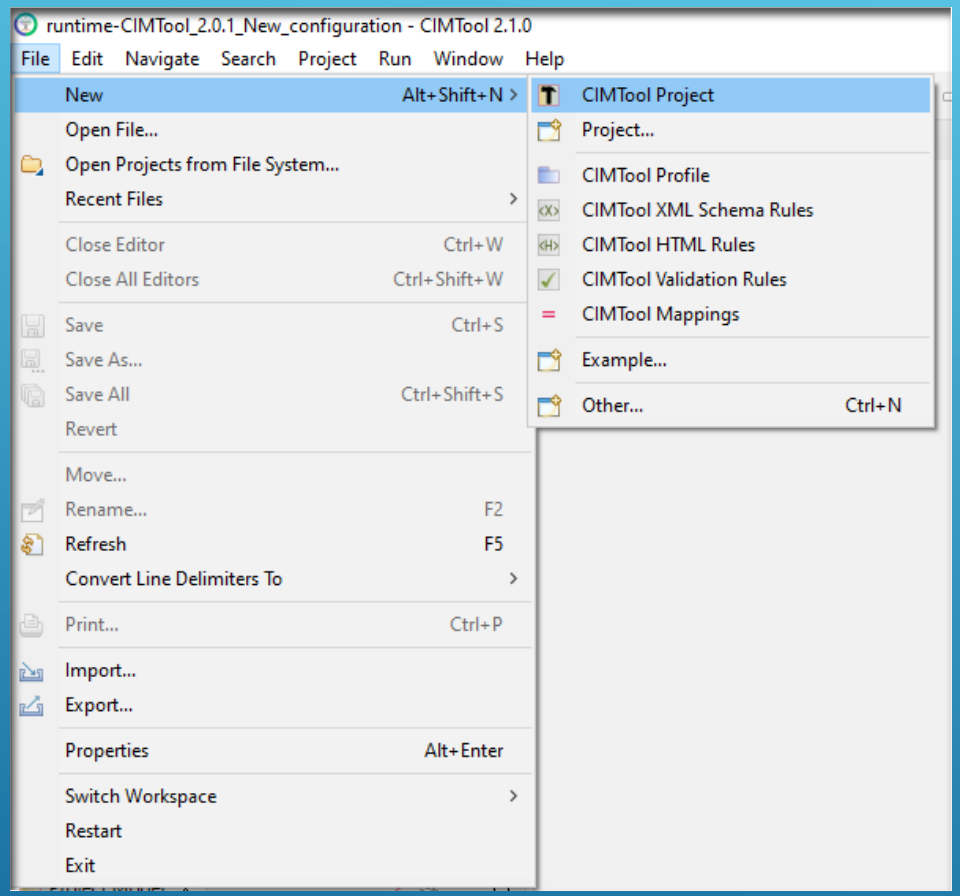
- *Open source* alat
  - *Site:* https://cimtool.ucaiug.io/
  - *Download*: https://github.com/cimug-org/CIMTool/releases

- Implementiran je kao plugin za Eclipse platformu
- Koristimo ga za dizajn CIM profila na bazi UML šeme modela date u XMI formatu.

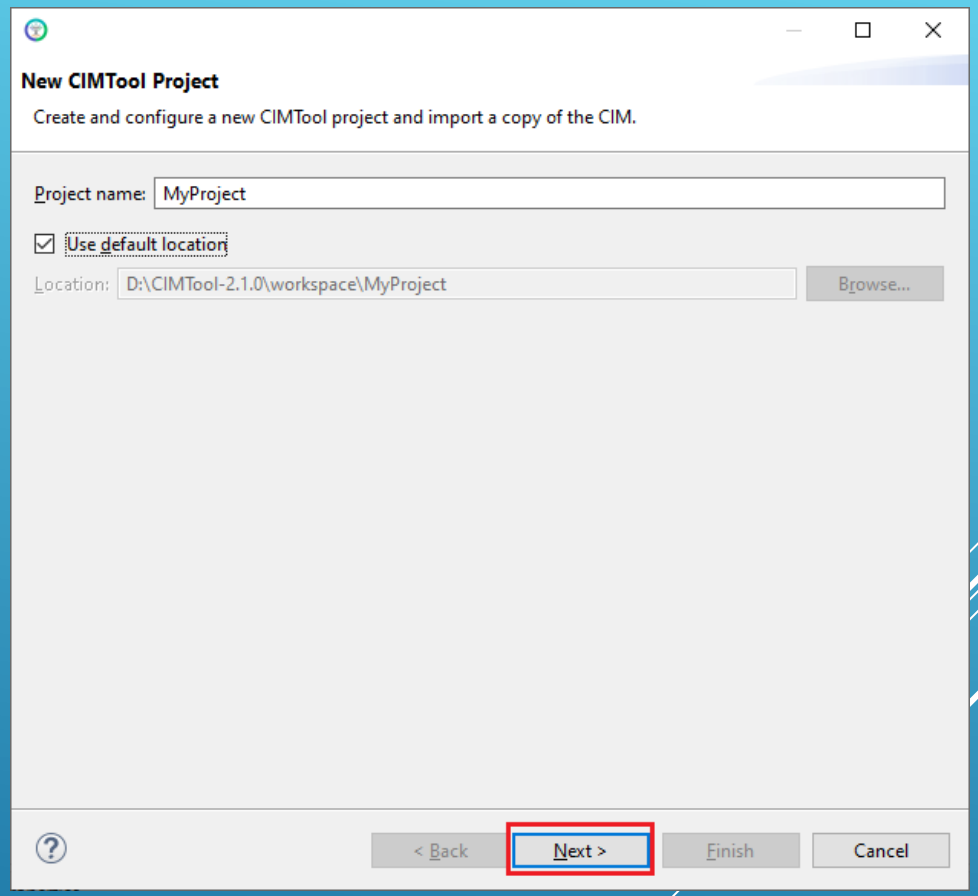
OSNOVNE OPERACIJE U CIMTOOL ALATU

- Kreiranje CIMTool projekta
- Učitavanje šeme modela date u XMI formatu
- Kreiranje CIM profila
  - Biranje formata zapisa CIM profila
    - HTML
    - RDFS
  - Dodavanje klase
  - Dodavanje atributa klase
  - Konfiguracija klasa i atributa u profilu
- Import postojećeg CIM profila u projekat
- Import postojećeg CIMTool projekta

CIMTOOL: KREIRANJE PROJEKTA

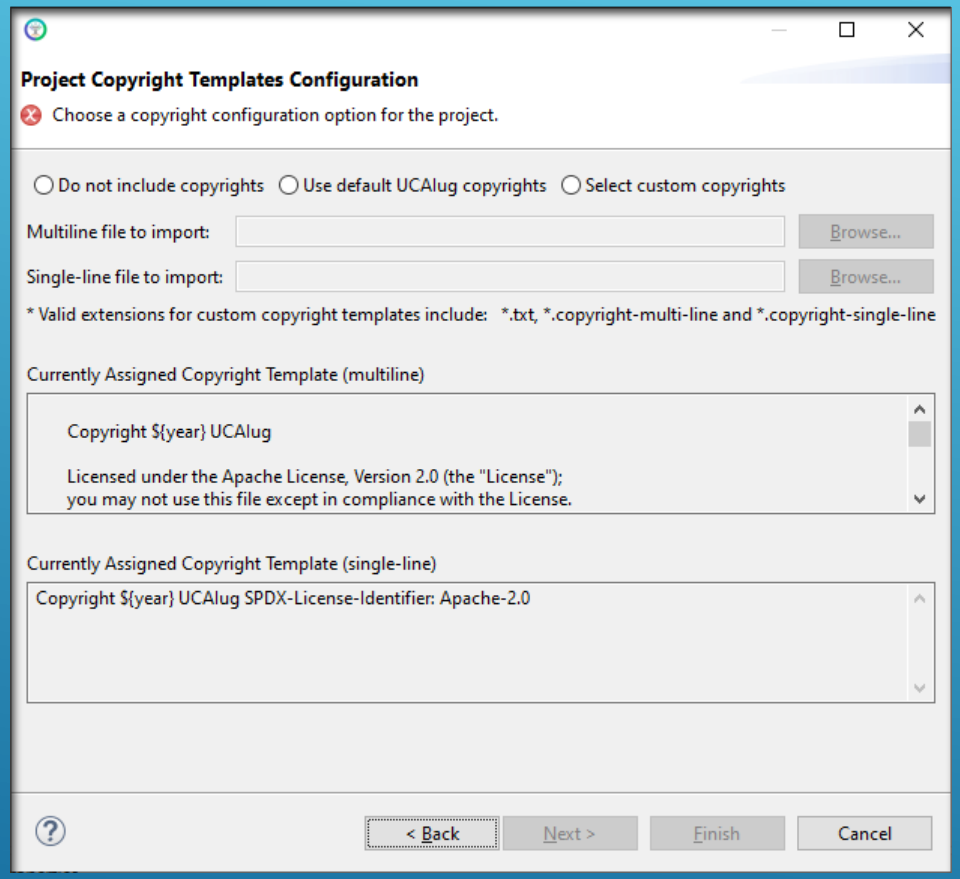


Korak 1

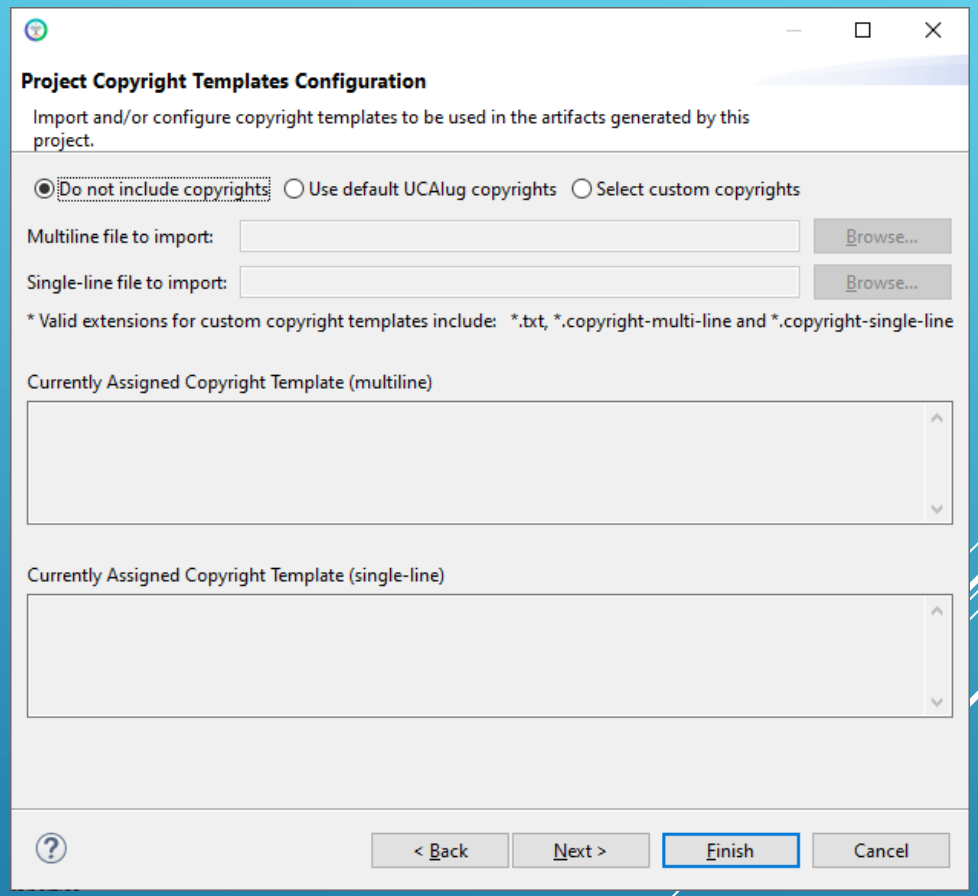


Korak 2

CIMTOOL: KREIRANJE PROJEKTA

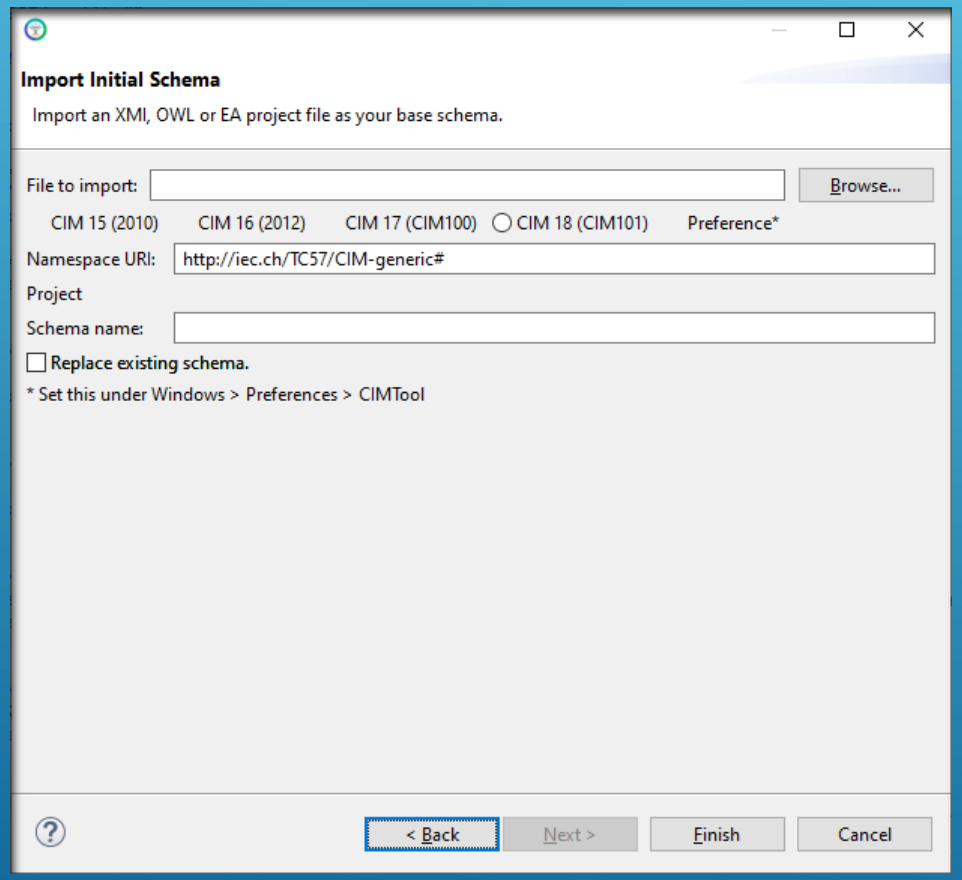


Korak 3

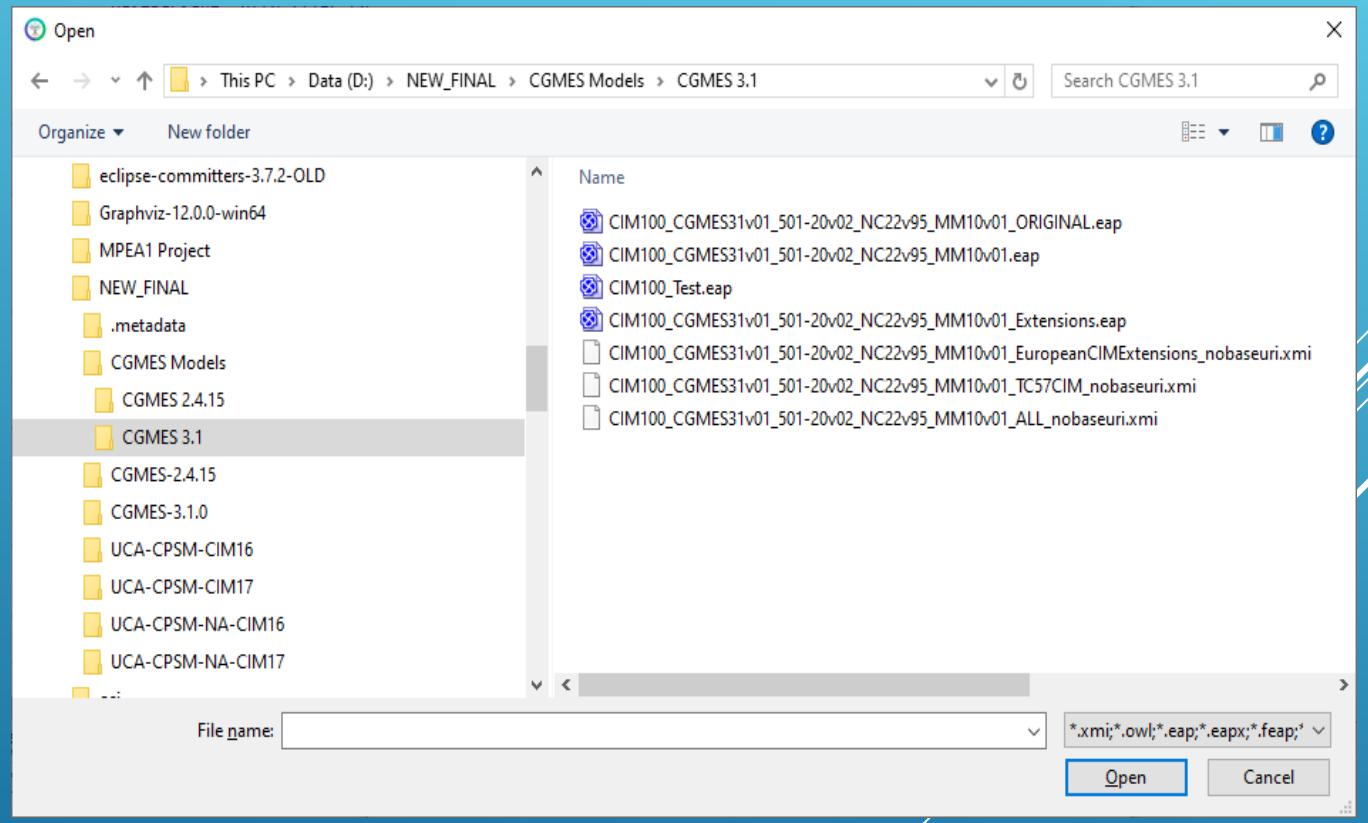


Korak 4

CIMTOOL: KREIRANJE PROJEKTA – UČITAVANJE ŠEME MODELA U XMI FORMATU

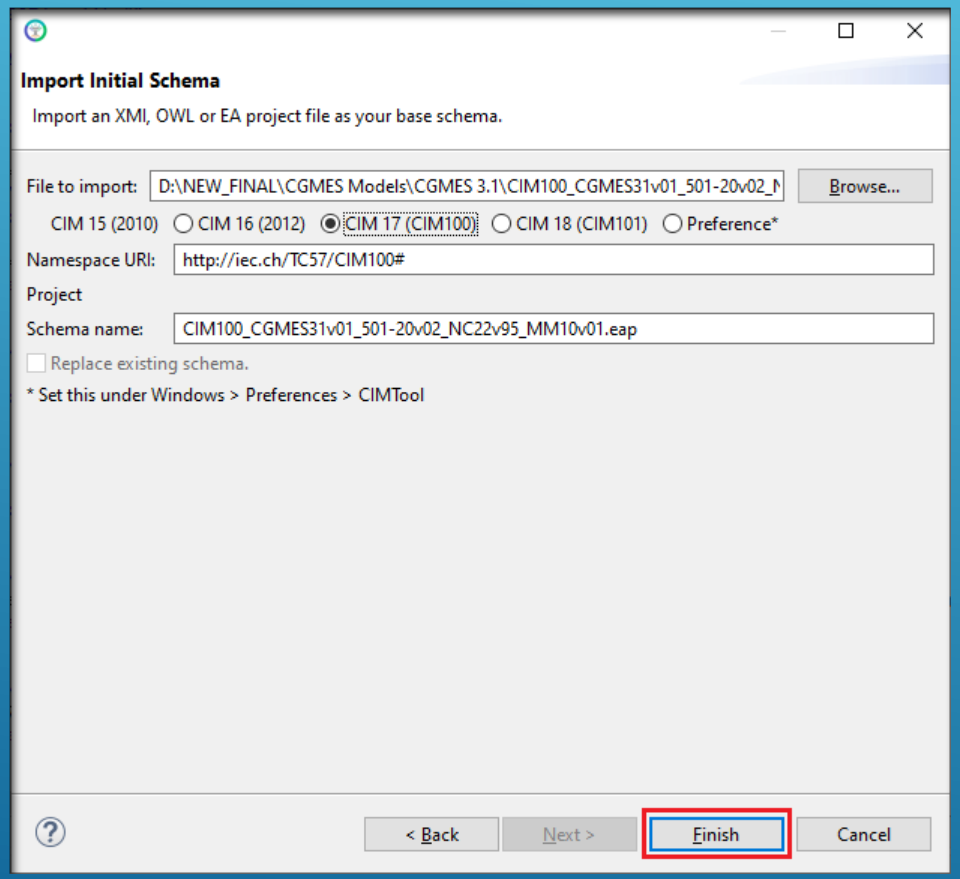


Korak 1

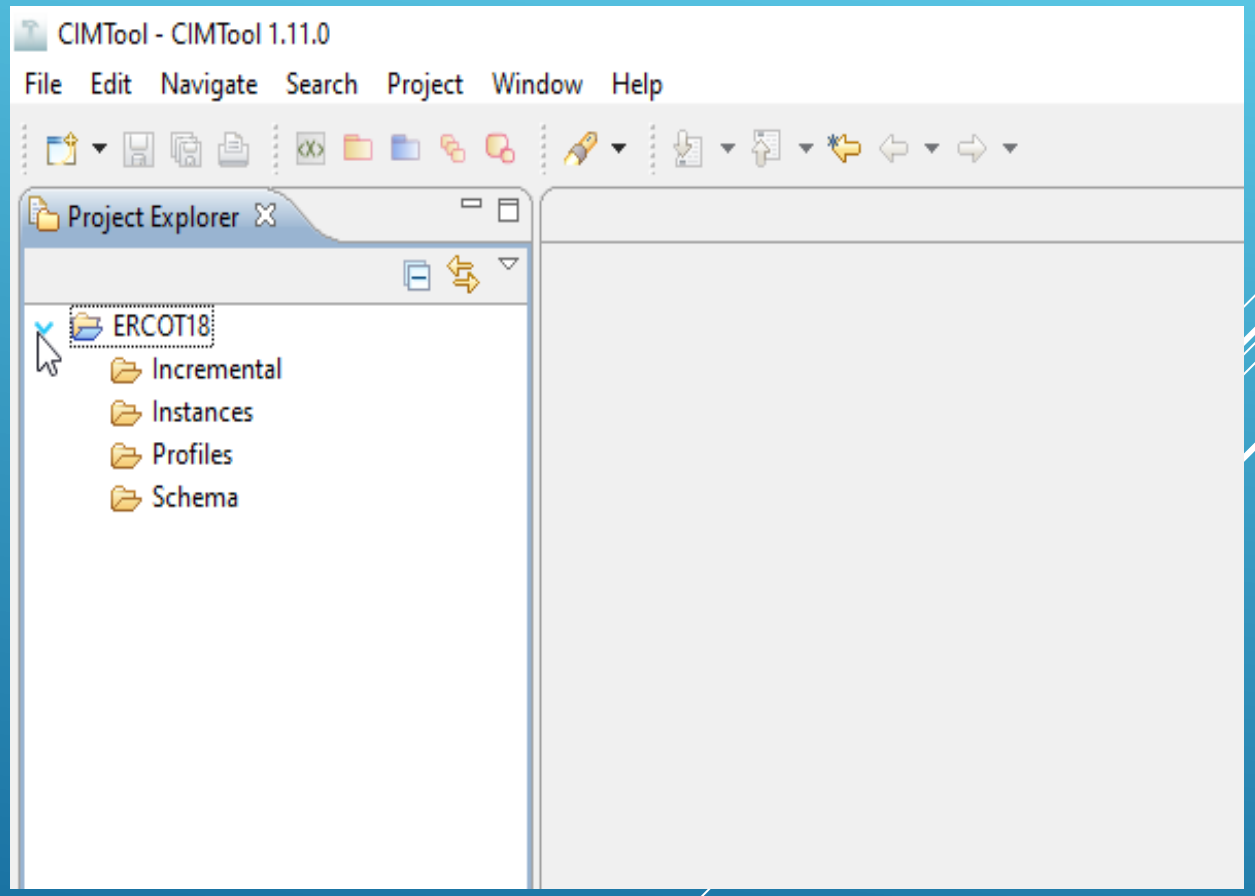


Korak 2

CIMTOOL: KREIRANJE PROJEKTA - UČITAVANJE ŠEME MODELA U XMI FORMATU

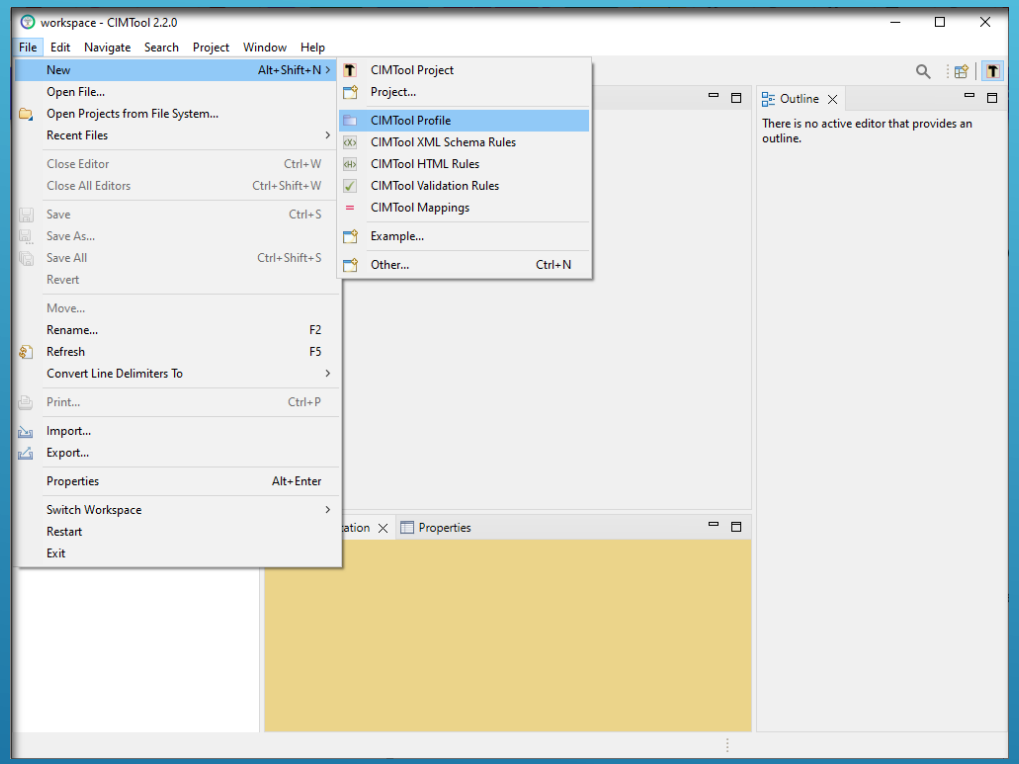


Korak 3

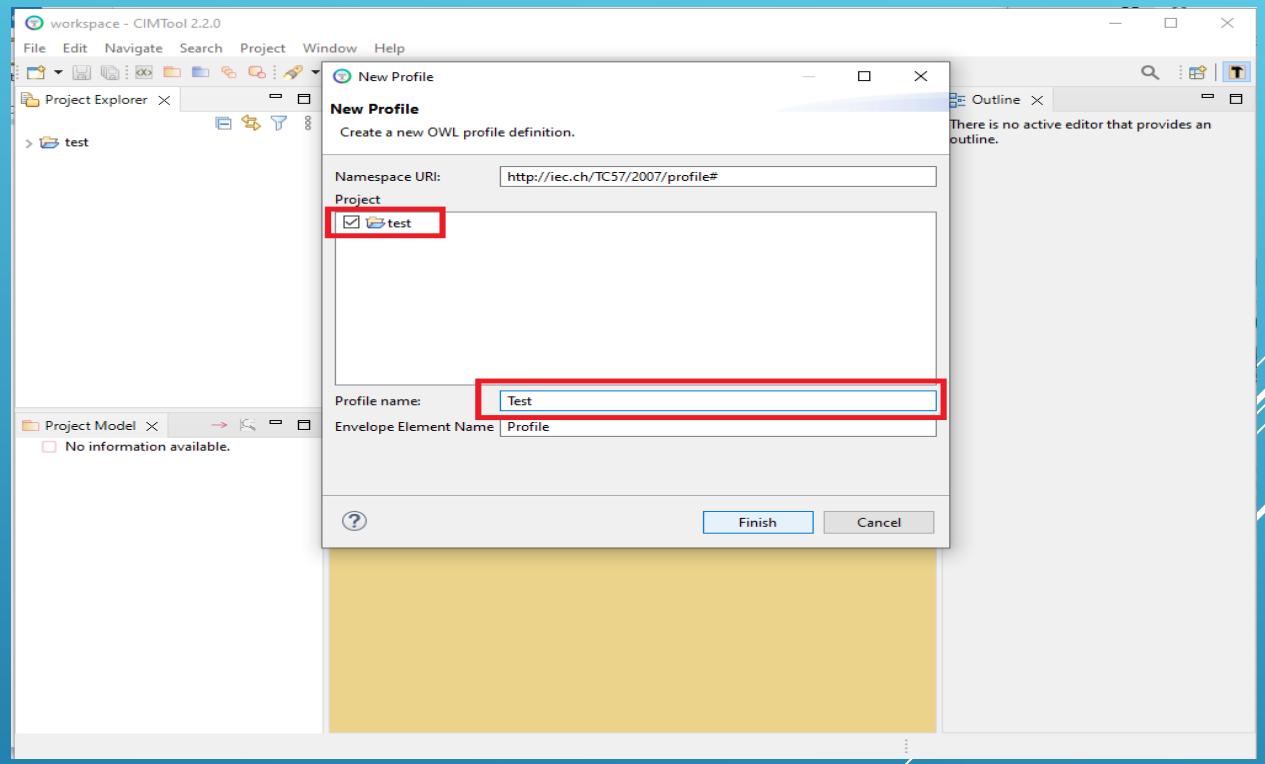


Rezultat u *Project Explorer*-u

CIMTOOL: KREIRANJE CIM PROFILA

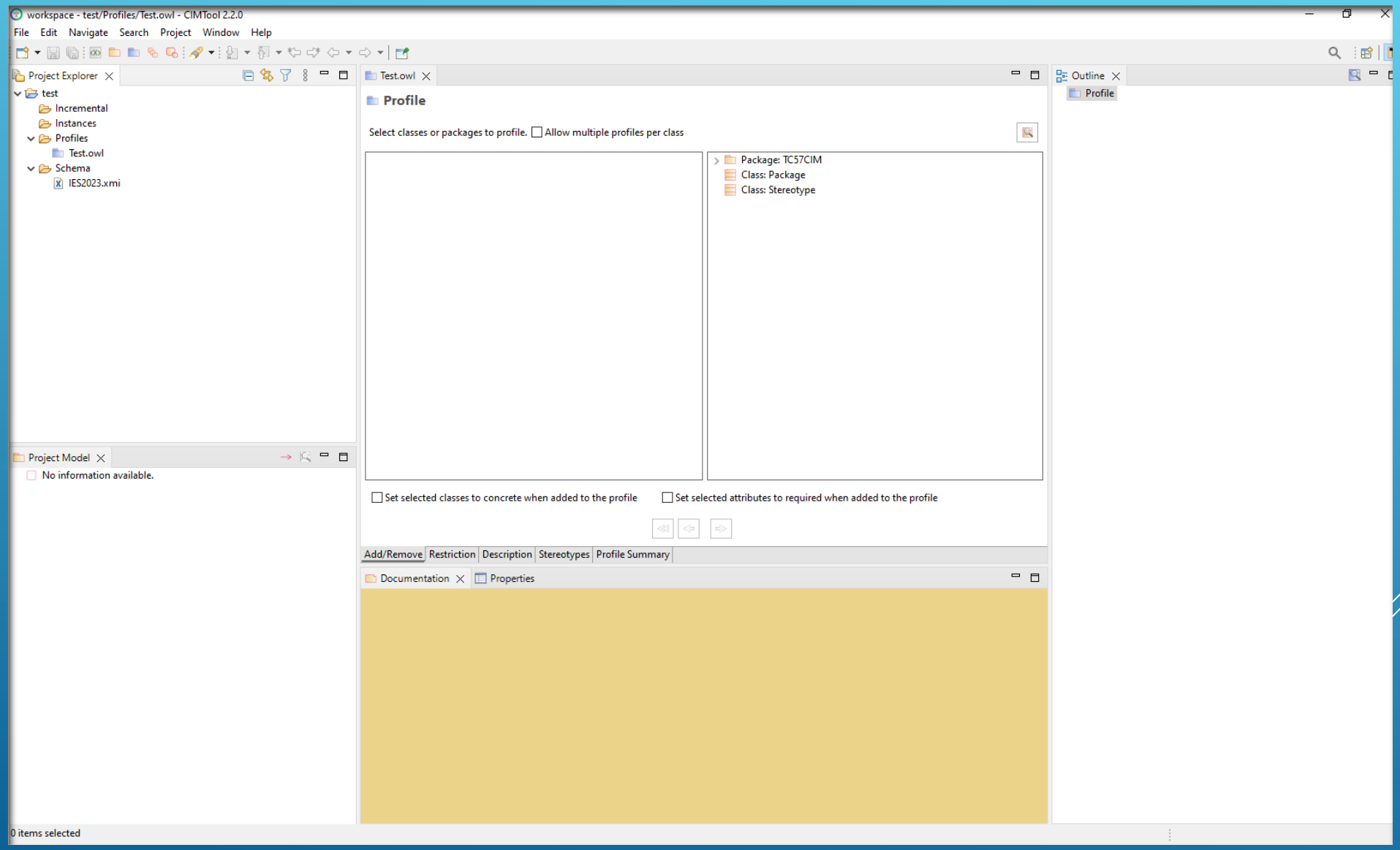


Korak 1



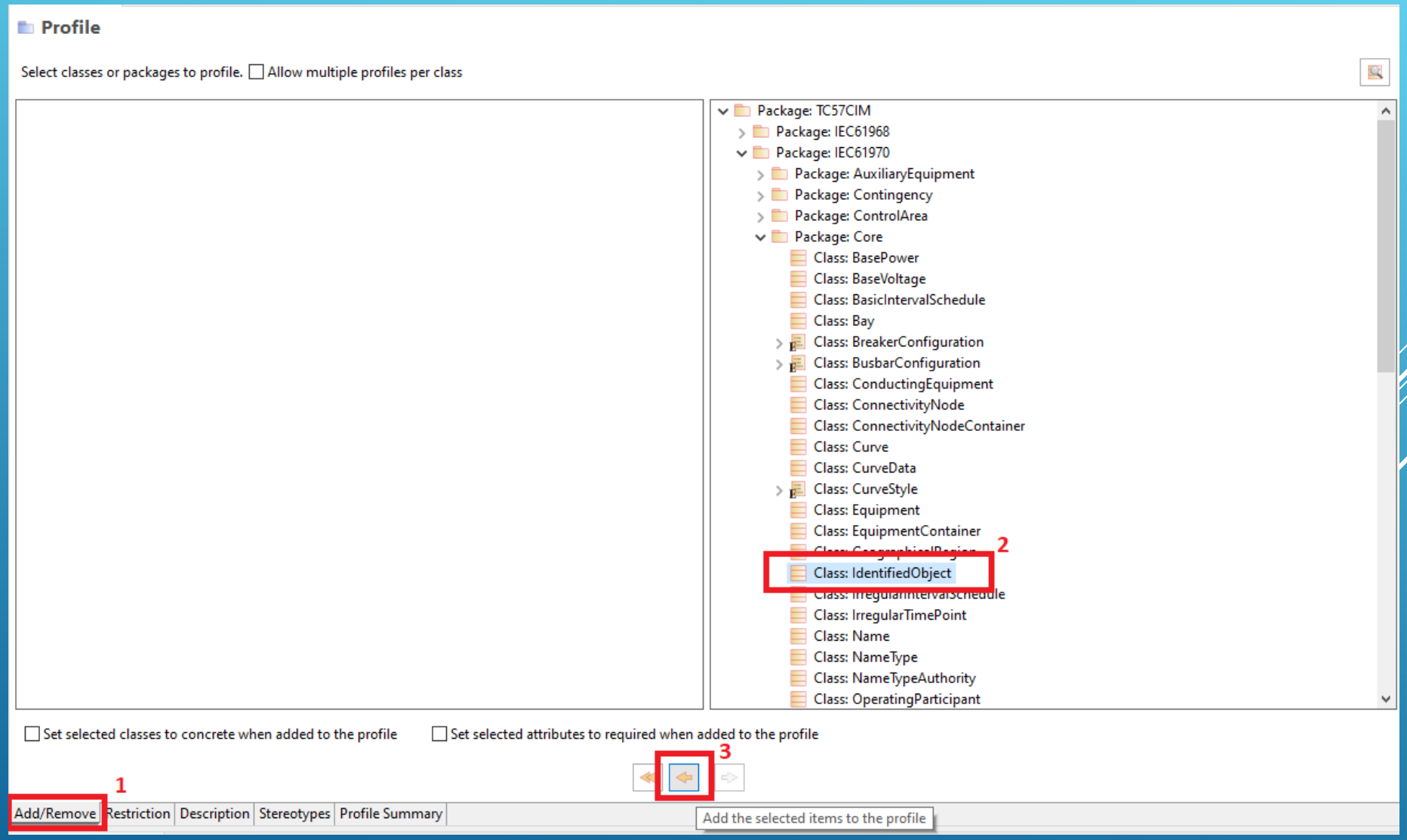
Korak 2

CIMTOOL: KREIRANJE CIM PROFILA



CIMTOOL: DODAVANJE KLASE

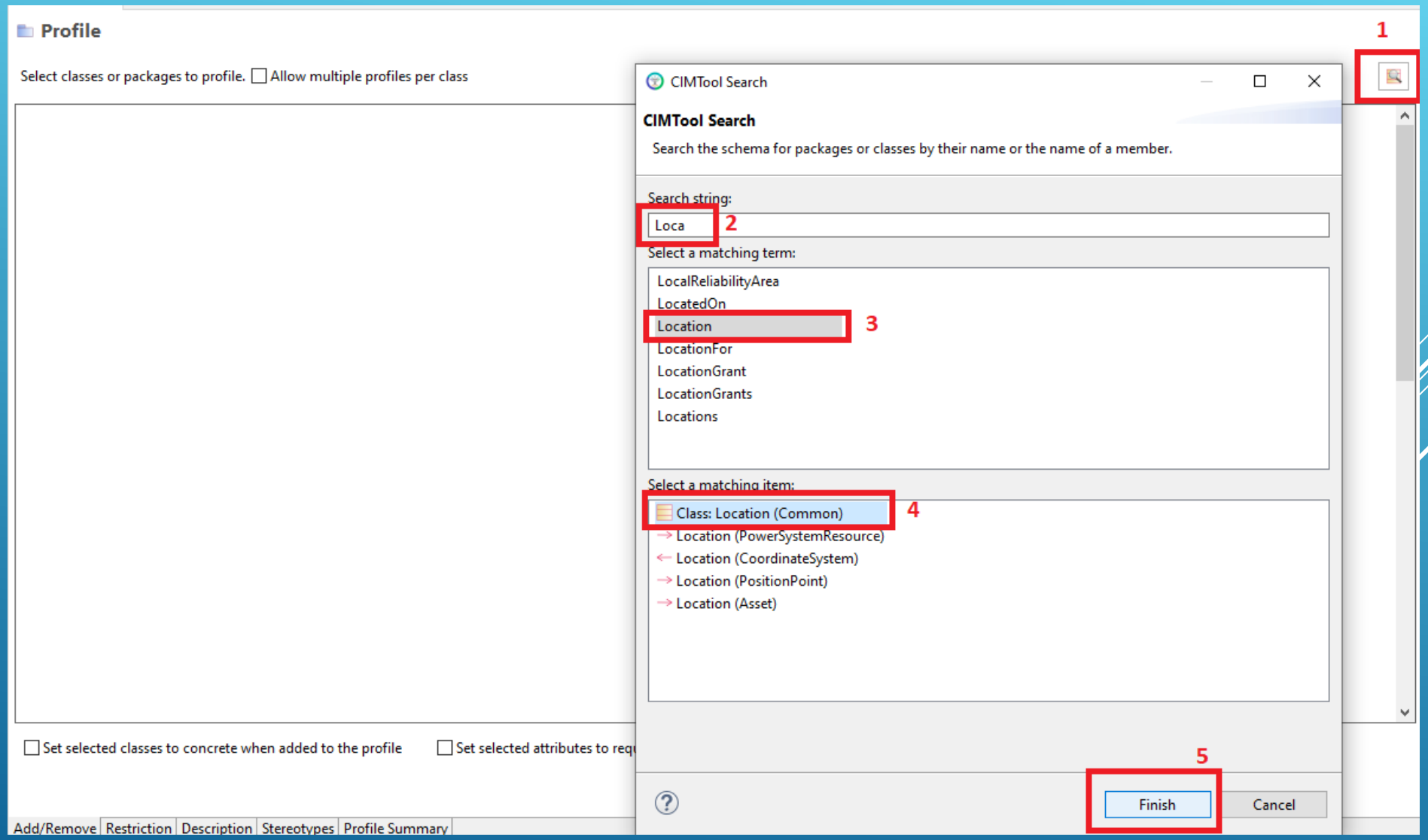
Prvi način:
odabirom iz liste



CIMTOOL: DODAVANJE KLASE

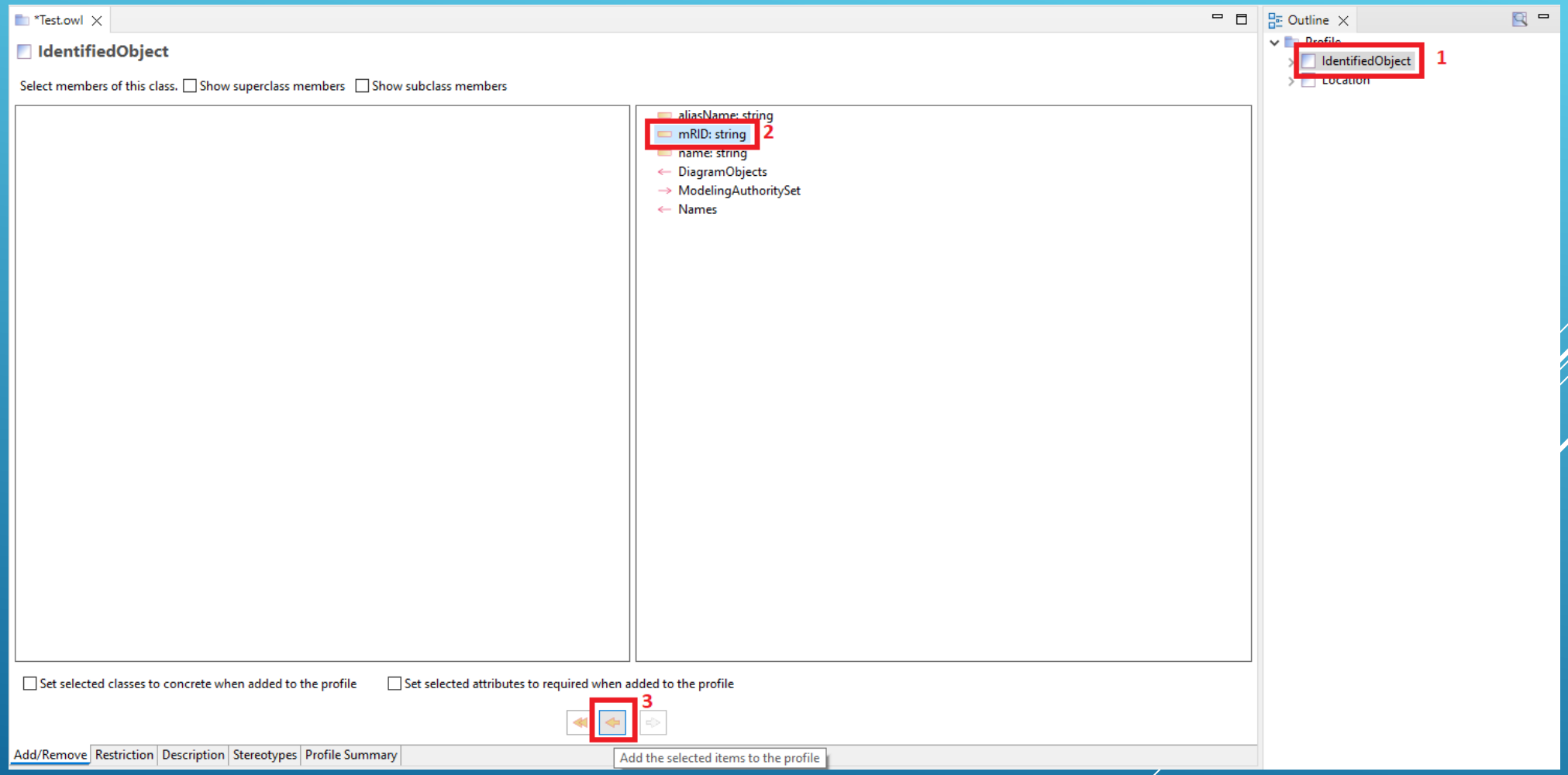
Drugi način:
Search Schema

VAŽNO! Pretraga je *case-sensitive*



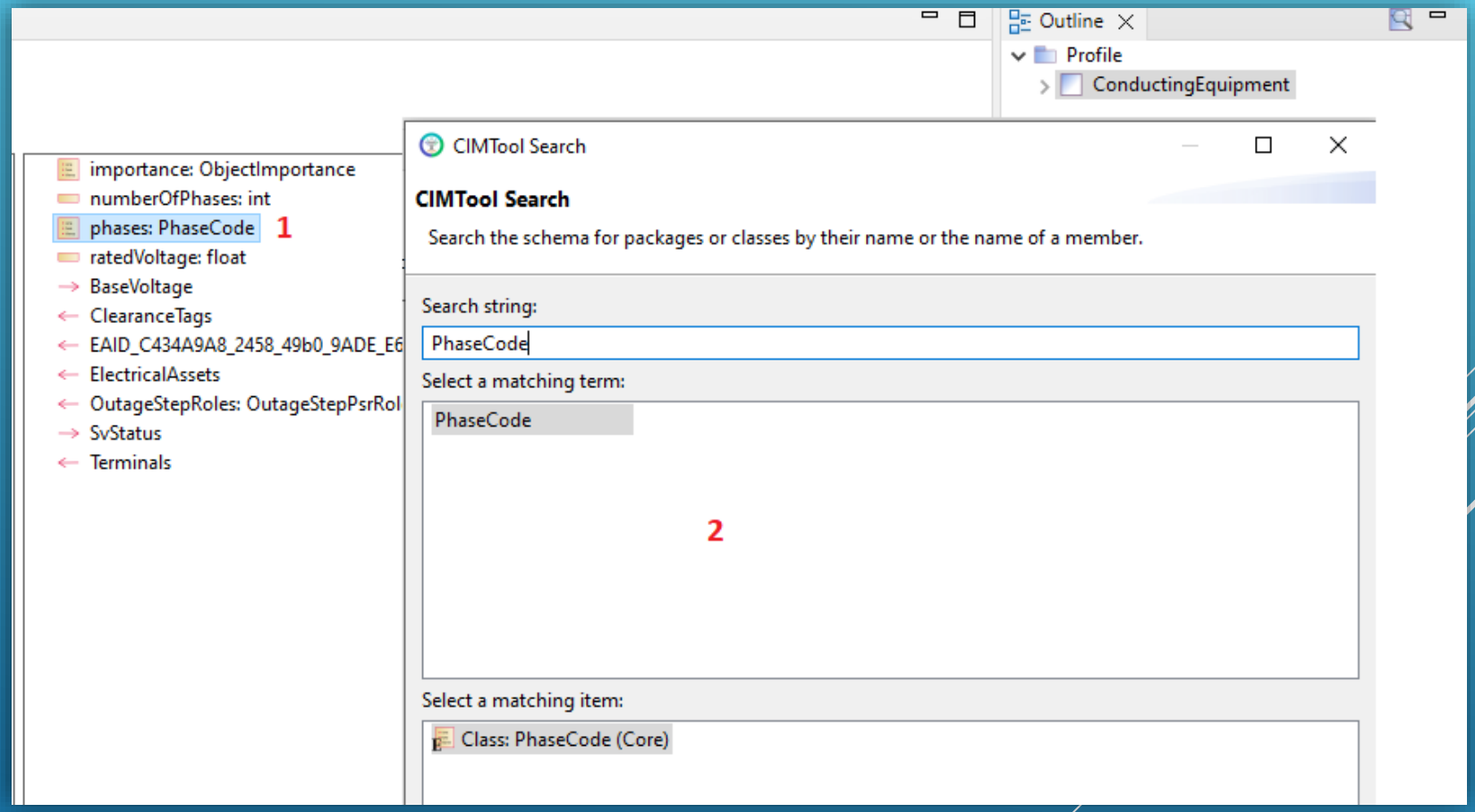
CIMTOOL: DODAVANJE ATRIBUTA

Koraci:



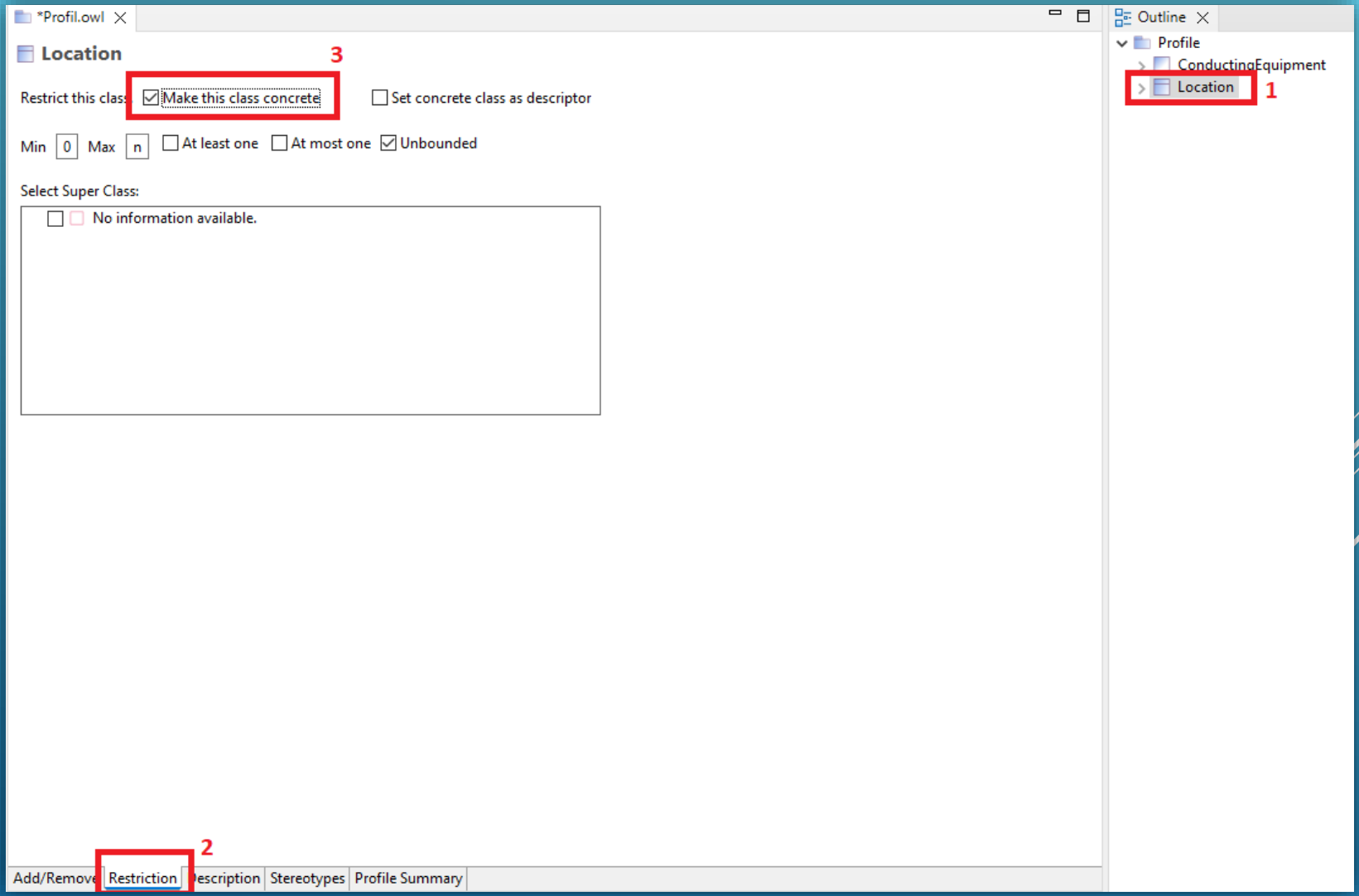
CIMTOOL: DODAVANJE ATRIBUTA TIPA ENUM

Pored dodavanja atributa u klasu, potrebno je dodati i samu enumeraciju i profil

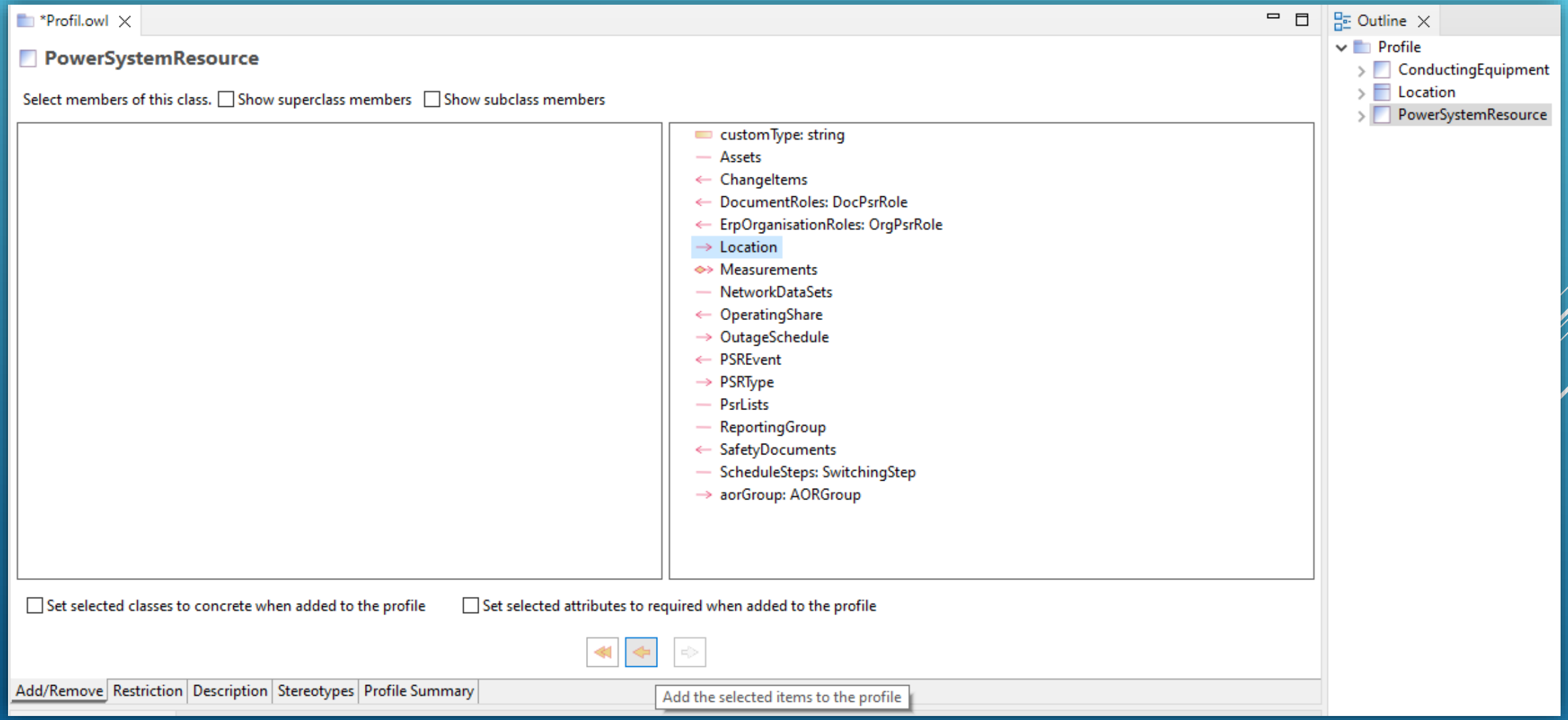


CIMTOOL: KONFIGURACIJA KLASE

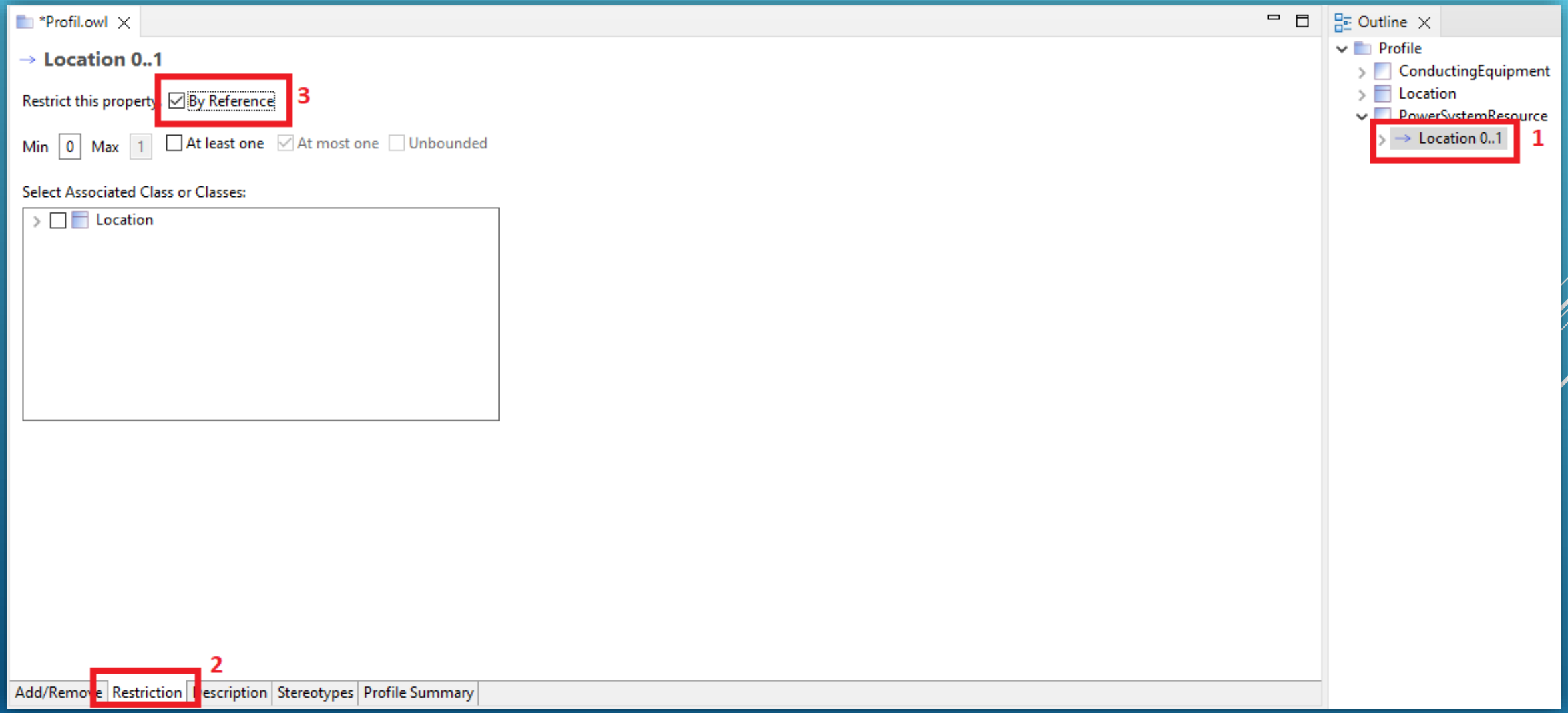
Abstract vs concrete



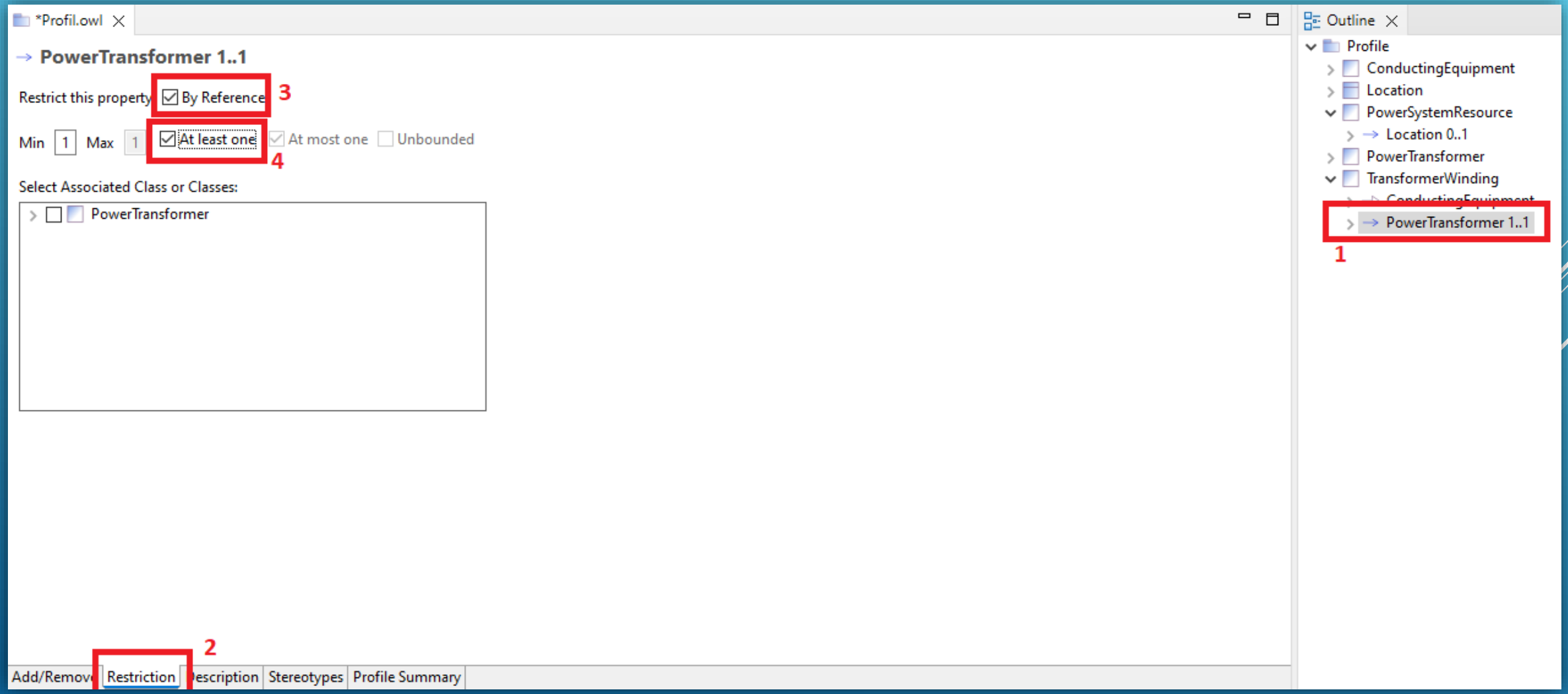
CIMTOOL: KONFIGURACIJA ATRIBUTA – BY REFERENCE



CIMTOOL: KONFIGURACIJA ATRIBUTA – BY REFERENCE



CIMTOOL: KONFIGURACIJA ATRIBUTA – KARDINALITETI



CIMTOOL: BIRANJE FORMATA ZAPISA CIM PROFILA

workspace - test/Profiles/Test.owl - CIMTool 2.2.0

File Edit 3 Navigate Search Project Window Help

Project Explorer

- test
  - Incremental
  - Instances
  - Profiles
    - Test.html
    - Test.legacy-rdfs
    - Test.owl
  - Schema
    - IES2023.xmi

4

Project Model

No information available.

Test.owl

Profile Summary

Location: /test/Profiles/Test.owl

Namespace: http://iec.ch/TC57/2007/profile# Edit

Reorganize and Repair | Manage XSLT Builders...

Build the following from this profile:

- [ ] Builder for cimantic-graphs (.cimantic-graphs.py)
- [ ] Builder for cimantic-graphs-init (.__init__.py)
- [ ] Builder for ecore (.ecore)
- [x] Builder for html (.html)
- [ ] Builder for jpa (.java)
- [ ] Builder for json-schema-draft-07 (.draft-07.schema.json)
- [ ] Builder for json-schema-draft-2020-12 (.draft-2020-12.schema.json)
- [x] Builder for legacy-rdfs (.legacy-rdfs)
- [ ] Builder for legacy-rdfs-augmented (.legacy-rdfs-augmented)
- [ ] Builder for linkml (.linkml.yaml)
- [ ] Builder for profile-doc-rtf (.rtf)
- [ ] Builder for puml-l2r (.l2r.puml)
- [ ] Builder for puml-t2b (.t2b.puml)
- [ ] Builder for rdfs-2020 (.rdfs-2020.rdfs)
- [ ] Builder for rdfs-501ed2-beta (.rdfs-501ed2-beta.rdfs)
- [ ] Builder for rdfs-augmented-cimcontextor (.augmented-cimcontextor.rdfs)
- [ ] Builder for scala (.scala)
- [ ] Builder for simple-flat-owl (.simple-flat-owl)
- [ ] Builder for simple-flat-owl-augmented (.simple-flat-owl-augmented)
- [ ] Builder for simple-owl (.simple-owl)

2

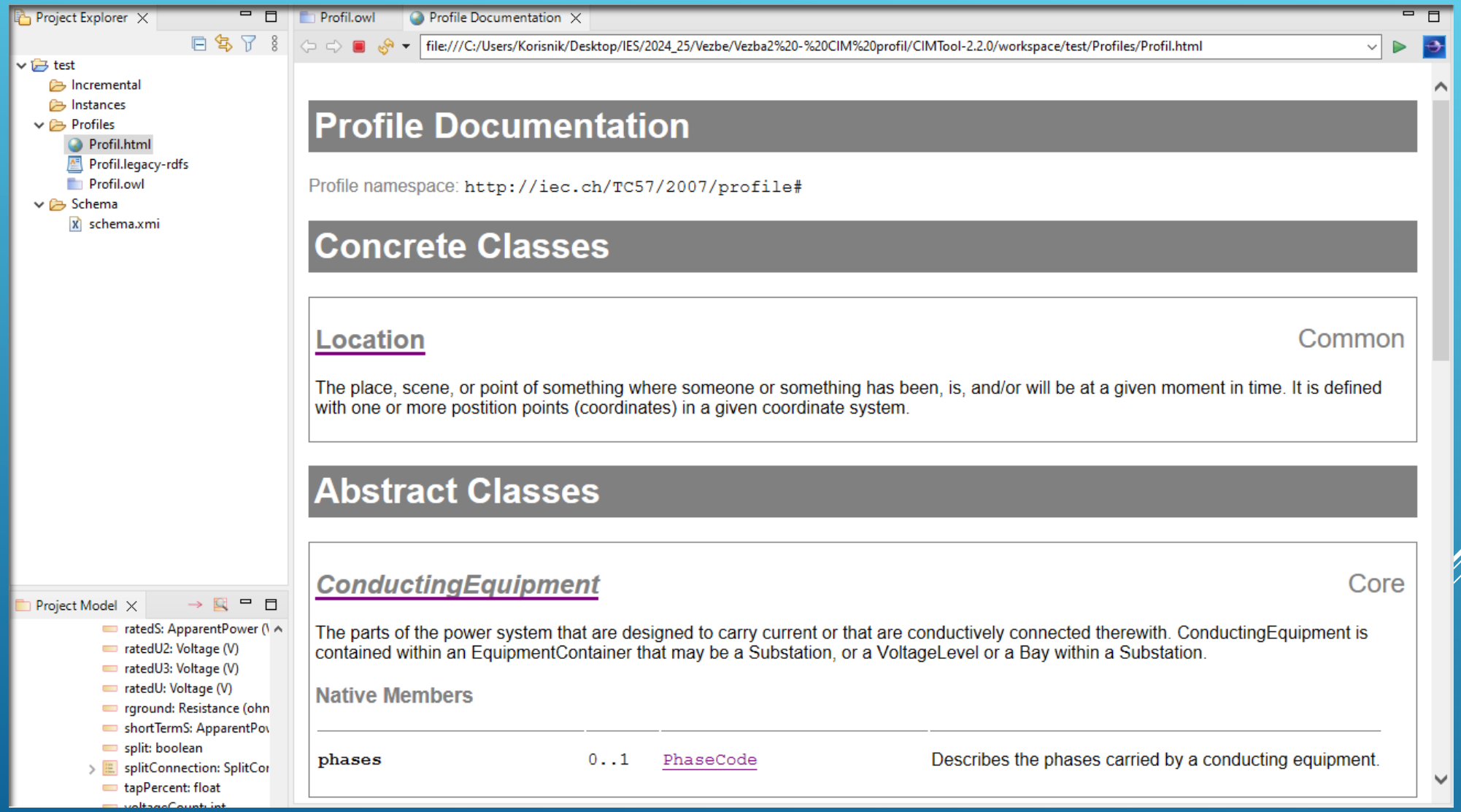
Add/Remove | Restriction | Description | Stereotypes | Profile Summary 1

Documentation | Properties

Outline

Profile

CIMTOOL: PREGLED HTML FORMATA ZAPISA



CIMTOOL: PREGLED RDFS FORMATA ZAPISA

Project Explorer

- test
  - Incremental
  - Instances
  - Profiles
    - Profil.html
    - Profil.legacy-rdfs
    - Profil.owl
  - Schema
    - schema.xmi

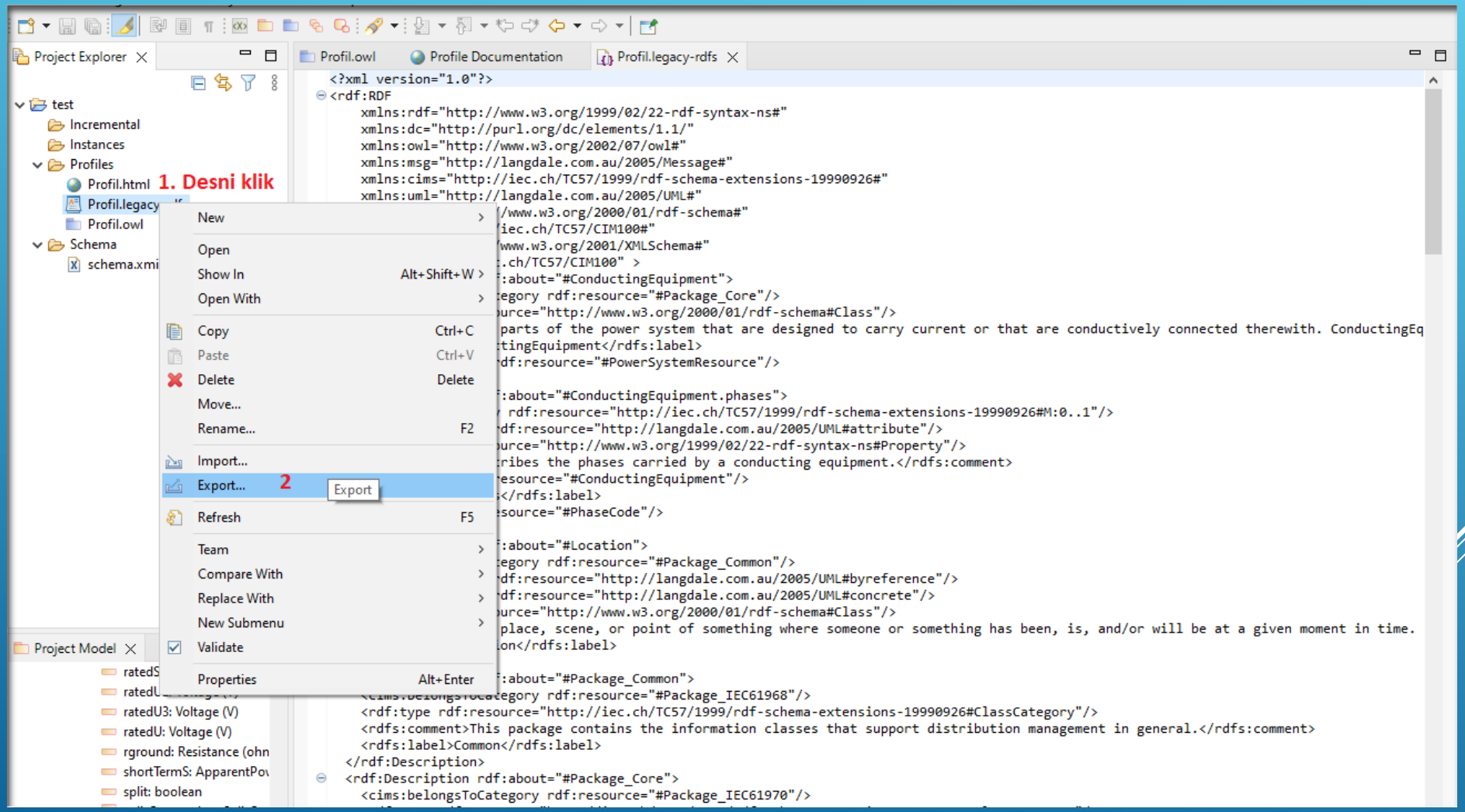
Profil.owl | Profile Documentation | Profil.legacy-rdfs

```
<?xml version="1.0"?>
<rdf:RDF
    xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
    xmlns:dc="http://purl.org/dc/elements/1.1/"
    xmlns:owl="http://www.w3.org/2002/07/owl#"
    xmlns:msg="http://langdale.com.au/2005/Message#"
    xmlns:cims="http://iec.ch/TC57/1999/rdf-schema-extensions-19990926#"
    xmlns:uml="http://langdale.com.au/2005/UML#"
    xmlns:rdfs="http://www.w3.org/2000/01/rdf-schema#"
    xmlns:j.0="http://iec.ch/TC57/CIM100#"
    xmlns:xsd="http://www.w3.org/2001/XMLSchema#"
  xml:base="http://iec.ch/TC57/CIM100" >
  <rdf:Description rdf:about="#ConductingEquipment">
    <cims:belongsToCategory rdf:resource="#Package_Core"/>
    <rdf:type rdf:resource="http://www.w3.org/2000/01/rdf-schema#Class"/>
    <rdfs:comment>The parts of the power system that are designed to carry current or that are conductively connected therewith. ConductingEq
    <rdfs:label>ConductingEquipment</rdfs:label>
    <rdfs:subClassOf rdf:resource="#PowerSystemResource"/>
  </rdf:Description>
  <rdf:Description rdf:about="#ConductingEquipment.phases">
    <cims:multiplicity rdf:resource="http://iec.ch/TC57/1999/rdf-schema-extensions-19990926#M:0..1"/>
    <cims:stereotype rdf:resource="http://langdale.com.au/2005/UML#attribute"/>
    <rdf:type rdf:resource="http://www.w3.org/1999/02/22-rdf-syntax-ns#Property"/>
    <rdfs:comment>Describes the phases carried by a conducting equipment.</rdfs:comment>
    <rdfs:domain rdf:resource="#ConductingEquipment"/>
    <rdfs:label>phases</rdfs:label>
    <rdfs:range rdf:resource="#PhaseCode"/>
  </rdf:Description>
  <rdf:Description rdf:about="#Location">
    <cims:belongsToCategory rdf:resource="#Package_Common"/>
    <cims:stereotype rdf:resource="http://langdale.com.au/2005/UML#byreference"/>
    <cims:stereotype rdf:resource="http://langdale.com.au/2005/UML#concrete"/>
    <rdf:type rdf:resource="http://www.w3.org/2000/01/rdf-schema#Class"/>
    <rdfs:comment>The place, scene, or point of something where someone or something has been, is, and/or will be at a given moment in time.
    <rdfs:label>Location</rdfs:label>
  </rdf:Description>
  <rdf:Description rdf:about="#Package_Common">
    <cims:belongsToCategory rdf:resource="#Package_IEC61968"/>
    <rdf:type rdf:resource="http://iec.ch/TC57/1999/rdf-schema-extensions-19990926#ClassCategory"/>
    <rdfs:comment>This package contains the information classes that support distribution management in general.</rdfs:comment>
    <rdfs:label>Common</rdfs:label>
  </rdf:Description>
  <rdf:Description rdf:about="#Package_Core">
    <cims:belongsToCategory rdf:resource="#Package_IEC61970"/>
    <rdf:type rdf:resource="http://iec.ch/TC57/1999/rdf-schema-extensions-19990926#ClassCategory"/>
```

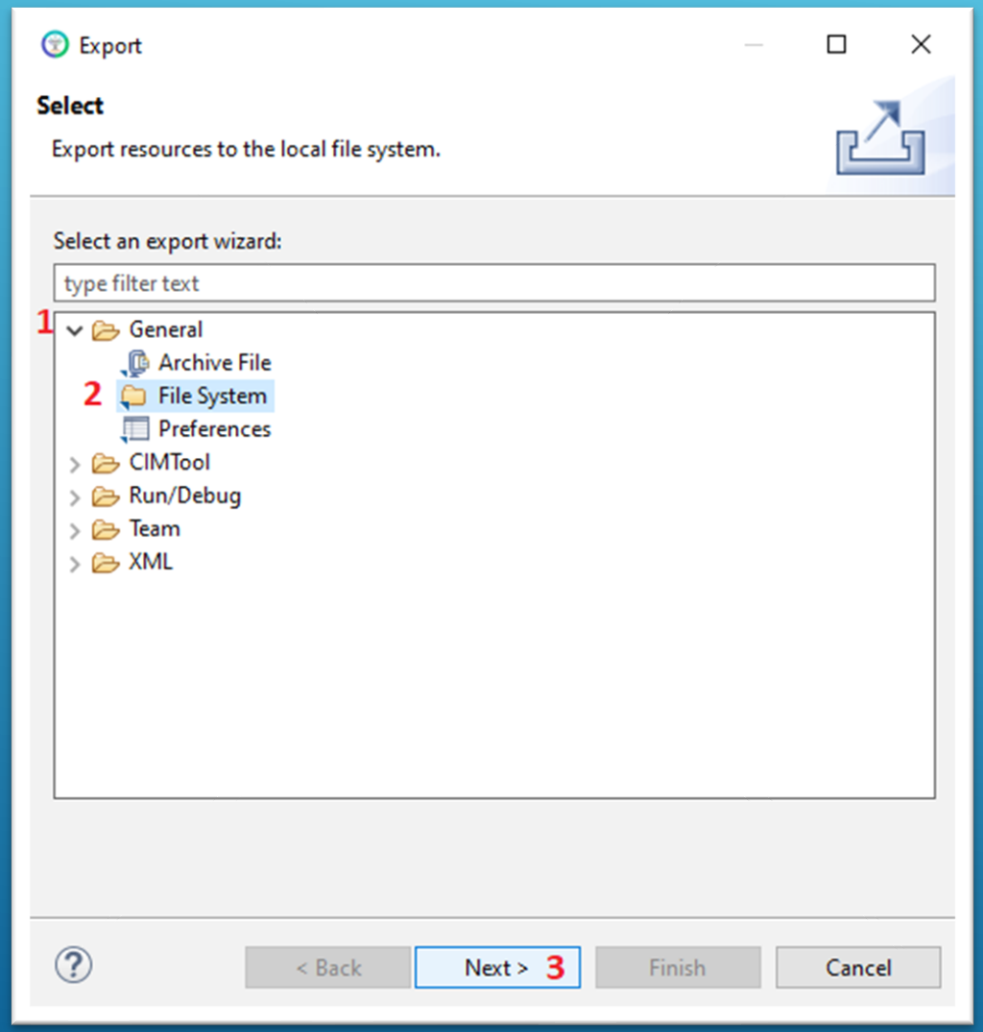
Project Model

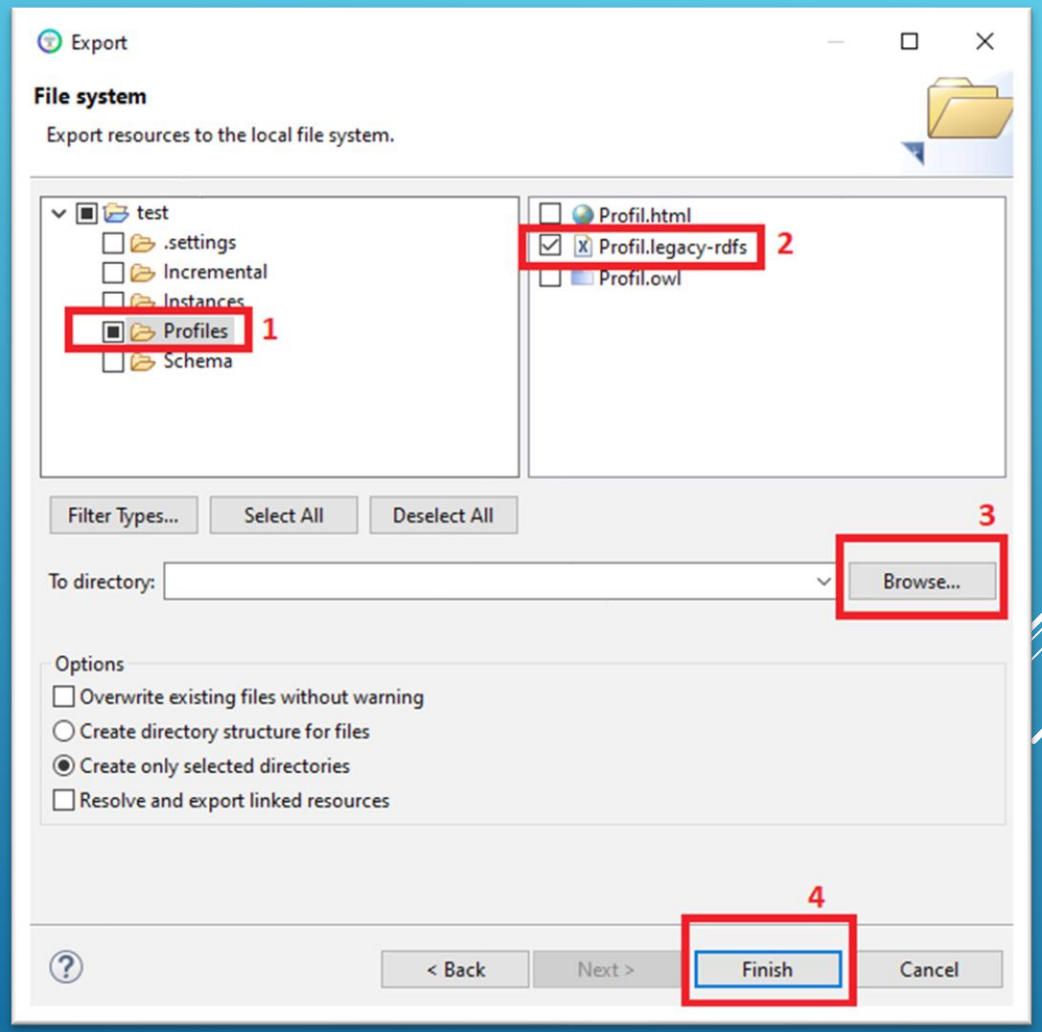
- ratedS: ApparentPower (\
- ratedU2: Voltage (V)
- ratedU3: Voltage (V)
- ratedU: Voltage (V)
- rground: Resistance (ohn
- shortTermS: ApparentPo
- split: boolean
- splitConnection: SplitCo
- tapPercent: float

CIMTOOL: EXPORT PROFILA U RDFS FORMATU



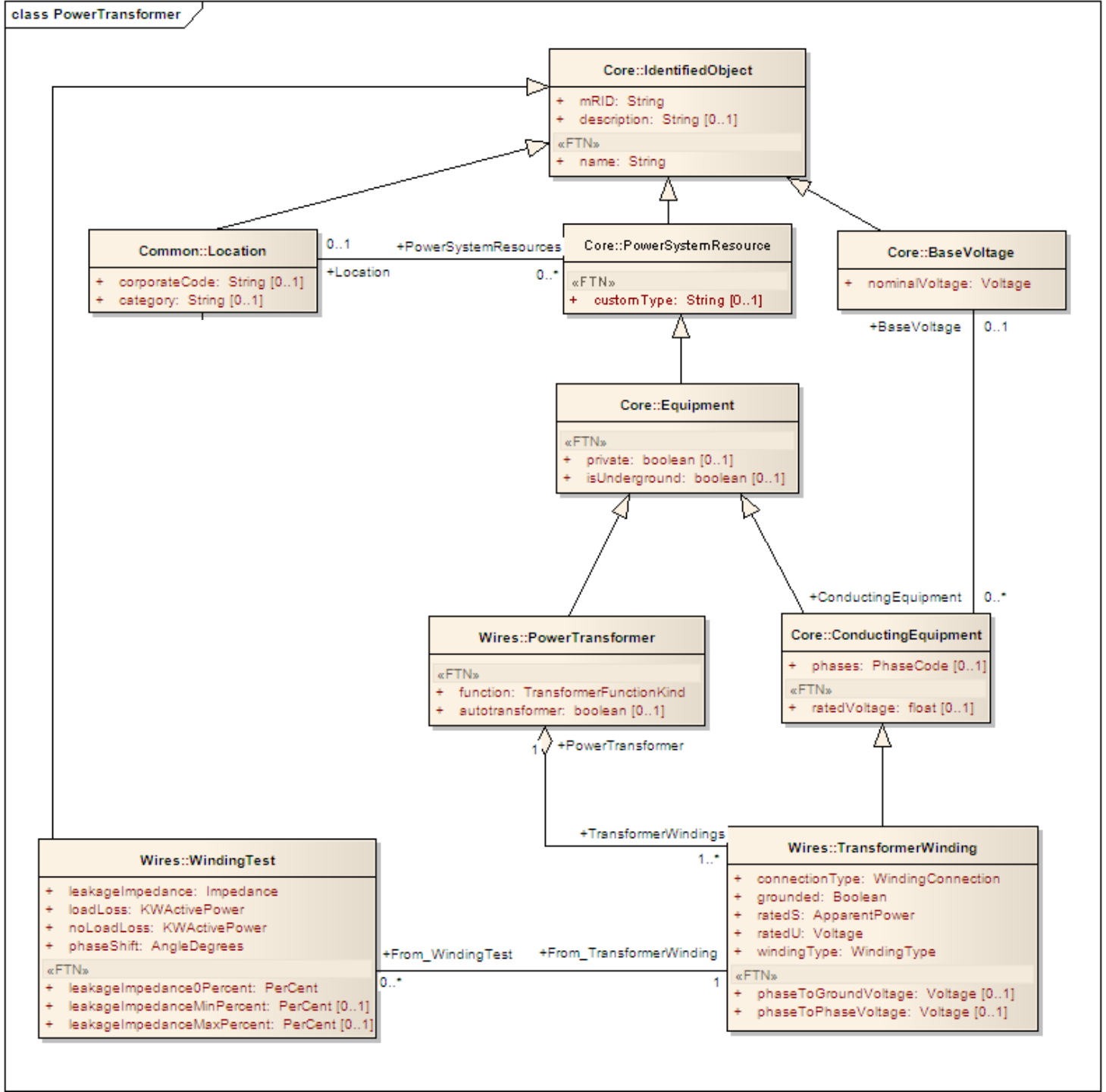
CIMTOOL: EXPORT PROFILA U RDFS FORMATU





CIMTOOL: DODATNE MOGUĆNOSTI

- Import postojećeg projekta
  - File → Import → General → Existing Projects into Workspace
- Import postojećeg profila
  - File → Import → CIMTool → Import Profile


class PowerTransformer
Core::IdentifiedObject
+ mRID: String
+ description: String [0..1]
«FTN»
+ name: String
Common::Location
+ corporateCode: String [0..1]
+ category: String [0..1]
0..1
+PowerSystemResources
+Location
0..*
Core::PowerSystemResource
«FTN»
+ customType: String [0..1]
Core::BaseVoltage
+ nominalVoltage: Voltage
+BaseVoltage
0..1
Core::Equipment
«FTN»
+ private: boolean [0..1]
+ isUnderground: boolean [0..1]
+ConductingEquipment
0..*
Wires::PowerTransformer
«FTN»
+ function: TransformerFunctionKind
+ autotransformer: boolean [0..1]
Core::ConductingEquipment
+ phases: PhaseCode [0..1]
«FTN»
+ ratedVoltage: float [0..1]
1
+PowerTransformer
+TransformerWindings
1..*
Wires::WindingTest
+ leakageImpedance: Impedance
+ loadLoss: KWActivePower
+ noLoadLoss: KWActivePower
+ phaseShift: AngleDegrees
«FTN»
+ leakageImpedance0Percent: PerCent
+ leakageImpedanceMinPercent: PerCent [0..1]
+ leakageImpedanceMaxPercent: PerCent [0..1]
+From_WindingTest
+From_TransformerWinding
0..*
1
Wires::TransformerWinding
+ connectionType: WindingConnection
+ grounded: Boolean
+ ratedS: ApparentPower
+ ratedU: Voltage
+ windingType: WindingType
«FTN»
+ phaseToGroundVoltage: Voltage [0..1]
+ phaseToPhaseVoltage: Voltage [0..1]

ZADATAK

1. Dodati sledeće klase u CIM profil:
   - IdentifiedObject
   - Location
   - BaseVoltage
   - PowerSystemResource
   - Equipment
   - PowerTransformer
   - ConductingEquipment
   - TransformerWinding
   - WindingTest
2. Za sve klase dodati atribute navedene na dijagramu sa prethodnog slajda. Konfigurisati kardinalitete po potrebi. Dodati enumeracije.
3. Konfigurisati konkretne klase.
4. Dodati atribute tipa referenca. Konfiguristi kardinalitete.